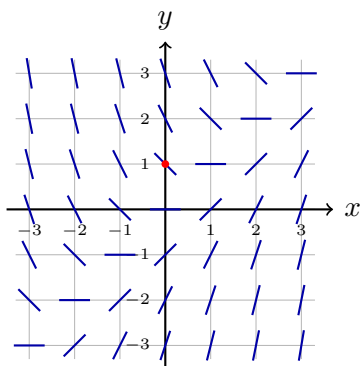


# Reading and Sketching Slope Fields

- Each printed field draws the slope  $\frac{dy}{dx}$  as a short segment at every lattice point. A solution curve is any curve that stays *tangent* to those segments — it never cuts across one.
- Sketch directly on the printed field. A dot marks the point your curve must pass through.
- Give exact values (fractions,  $e$ ) unless a decimal is requested.

1. The field below is for  $\frac{dy}{dx} = x - y$ . The dot marks  $(0, 1)$ .



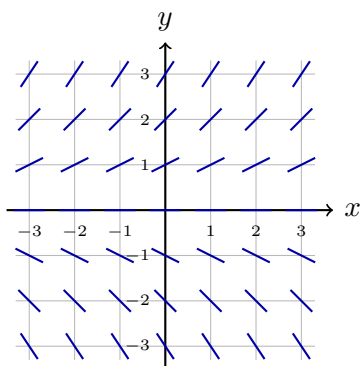
- (a) Find the slope of a solution curve at  $(2, -1)$ .

Answer: \_\_\_\_\_

- (b) Sketch the solution curve through the marked point.

Answer: \_\_\_\_\_

2. The field below is for  $\frac{dy}{dx} = \frac{y}{2}$ .



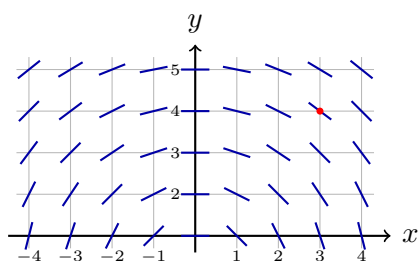
- (a) Find the slope of a solution curve at  $(1, 4)$ .

Answer: \_\_\_\_\_

- (b) Find every value of  $y$  at which the printed segments are horizontal.

Answer: \_\_\_\_\_

3. The field below is for  $\frac{dy}{dx} = -\frac{x}{y}$ , drawn only where  $y > 0$ . The dot marks  $(3, 4)$ .



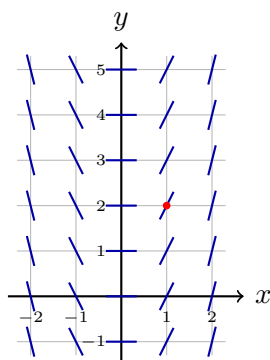
- (a) Find the slope of a solution curve at the marked point.

Answer: \_\_\_\_\_

- (b) Sketch the solution curve through it and describe its shape.

Answer: \_\_\_\_\_

4. The field below is for  $\frac{dy}{dx} = 2x$ . The dot marks  $(1, 2)$ .



- (a) Antidifferentiate to find the general solution  $y = \dots + C$ .

Answer: \_\_\_\_\_

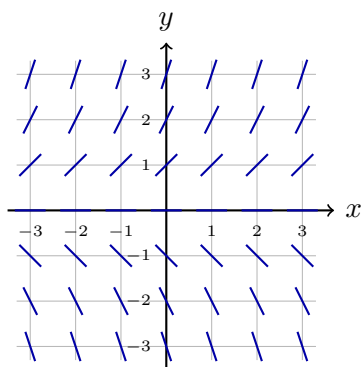
- (b) Find  $C$  for the solution through the marked point.

Answer: \_\_\_\_\_

- (c) Sketch that particular solution on the field.

Answer: \_\_\_\_\_

5. The field below is for one of  $\frac{dy}{dx} = x$  and  $\frac{dy}{dx} = y$ .



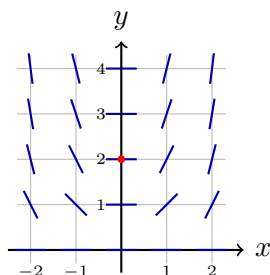
- (a) For  $\frac{dy}{dx} = y$ , find every  $y$  giving a horizontal segment.

Answer: \_\_\_\_\_

- (b) Explain which of the two equations the printed field shows.

Answer: \_\_\_\_\_

6. The field below is for  $\frac{dy}{dx} = xy$ . The dot marks  $(0, 2)$ . Separating gives  $\int \frac{dy}{y} = \int x \, dx$ , and the solution through that point is  $y = 2e^{x^2/2}$ .



- (a) Find the antiderivative  $\int x \, dx$ .

Answer: \_\_\_\_\_

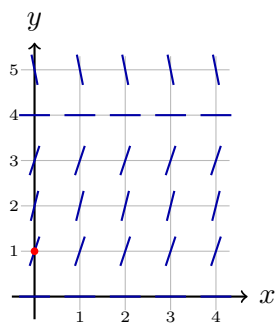
- (b) Find the exact value of  $y$  at  $x = 2$ .

Answer: \_\_\_\_\_

- (c) Sketch the particular solution as far as the grid allows.

Answer: \_\_\_\_\_

7. The field below is for  $\frac{dy}{dx} = y(4 - y)$ . The dot marks  $(0, 1)$ .



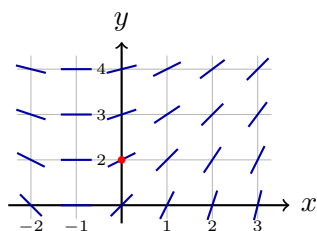
- (a) Find every value of  $y$  at which the segments are horizontal.

Answer: \_\_\_\_\_

- (b) Describe what happens to the solution through the marked point as  $x$  grows.

Answer: \_\_\_\_\_

8. The field below is for  $\frac{dy}{dx} = \frac{x+1}{y}$ , drawn where  $y > 0$ . The dot marks  $(0, 2)$ , and the solution through it is  $y = \sqrt{x^2 + 2x + 4}$ .



- (a) Find the slope of a solution curve at  $(1, 2)$ .

Answer: \_\_\_\_\_

- (b) Find  $y$  at  $x = 2$ , to three decimal places.

Answer: \_\_\_\_\_

- (c) Sketch the particular solution and name one way it agrees with the field.

Answer: \_\_\_\_\_